

Use Matlab for all three problems

26. A numerically controlled machine tool positioning servomechanism has a normalized and scaled transfer function given by

$$G(s) = \frac{1}{s(s+1)}.$$

Performance specifications of the system in the unity feedback configuration of Fig. 5.60 are satisfied if the closed-loop poles are located at $s = -1 \pm j\sqrt{3}$.

- (a) Show that this specification cannot be achieved by choosing proportional control alone, $D(s) = k_p$.
- (b) Design a lead compensator $D(s) = K \frac{s+z}{s+p}$ that will meet the specification.

27. A servomechanism position control has the plant transfer function

$$G(s) = \frac{10}{s(s+1)(s+10)}.$$

You are to design a series compensation transfer function $D(s)$ in the unity feedback configuration to meet the following closed-loop specifications:

- The response to a reference step input is to have no more than 16% overshoot.
 - The response to a reference step input is to have a rise time of no more than 0.4 sec.
 - The steady-state error to a unit ramp at the reference input must be less than 0.02
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- (a) Design a lead compensation that will cause the system to meet the dynamic response specifications.
 - (b) If $D(s)$ is proportional control, $D(s) = k_p$, what is the velocity constant K_v ?
 - (c) Design a lag compensation to be used in series with the lead you have designed to cause the system to meet the steady-state error specification.
 - (d) Give the MATLAB plot of the root locus of your final design.
 - (e) Give the MATLAB response of your final design to a reference step .

28. Assume the closed-loop system of Fig. 5.60 has a feed forward transfer function $G(s)$ given by

$$G(s) = \frac{1}{s(s+2)}.$$

Design a lag compensation so that the dominant poles of the closed-loop system are located at $s = -1 \pm j$ and the steady-state error to a unit ramp input is less than 0.2.

Hint: It says "Design a lag compensation ... the dominant poles ...". Usually by just using a lag compensator, we can only improve the steady state error but not the transient. We usually need a lead compensator to improve the transient behavior. But for this problem, you can just select the gain K to first meet the transient specification (closed poles at $-1 \pm j$) and there is no need to introduce a lead compensator to do so (because $-1 \pm j$ happens to be on the root locus ---- there is no need to further bend the locus to the upper left by using a lead compensator). After you select K , just design the lag compensator to reduce the steady state error.

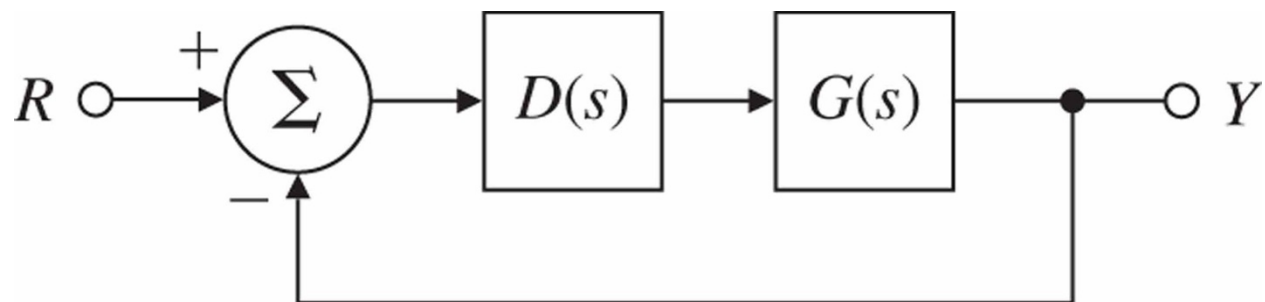


Fig. 5.60